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Receiving credit: On delegation and responsibility[☆]

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ABSTRACT

Evidence has shown that blame for a “bad” decision can be shifted by delegating the decision to someone else. We conduct experiments using a four-player delegated dictator game to examine whether the reverse is also true: does one receive credit for taking a “good” decision as compared to delegating the decision to someone else? Our results indicate that the answer is affirmative. A person receives higher rewards when she makes a fair decision herself than when a delegate does. This indicates that responsibility attribution is a double-edged sword that applies to both bad and good outcomes.

1. Introduction

The notion that the responsibility for unattractive decisions can be shifted to others has a long tradition. “Princes ought to leave affairs of reproach to the management of others,” Machiavelli (2008) wrote in *The Prince* (Chapter XIX). The effectiveness of blame-shifting is illustrated, *inter alia*, in economic experiments by Bartling and Fischbacher (2012). They show that decision-makers can avoid being punished for an unfair decision by delegating the decision to another person. Machiavelli (2008) also proposed that favorable decisions should *not* be delegated: “Princes ought to keep affairs of grace in their own hands”. In the current paper, we study the effectiveness of this latter strategy. Do decision-makers receive more credit when making favorable decisions themselves rather than delegating them to others?

To address this question, we conduct an economic experiment with a four-player delegated dictator game as implemented by Bartling and Fischbacher (2012). The first player (the dictator) can decide between an equal (fair) and an unequal (unfair) allocation that determines the payoffs of four players. Instead of making the decision herself, she can also pass the decision to the second player (the delegate), who then chooses between the two allocations. The monetary payoffs of the first and second players are perfectly aligned. The fair allocation divides the total amount equally among all four players, whereas the unfair allocation gives higher payoffs to the dictator and the delegate but lower payoffs to the other two players (the recipients). In Bartling and Fischbacher (2012), the recipients can decide to punish the dictator or the delegate, or both. In our experiments, we allow the recipients to choose to reward, rather than punish, the dictator or the delegate, or both. This allows us to explore whether credit-taking is symmetric to blame avoidance.

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Bartling and Fischbacher (2012) propose a model of responsibility attribution to explain how the recipients punish the dictator, the delegate, or both, in case the unfair allocation is chosen and how this depends on who made the final decision. We show that the model naturally extends to the case of rewards. If the fair allocation is chosen, responsibility is attributed in proportion to which player – dictator or delegate – contributed more to this outcome occurring. Under some conditions, the model predicts that the dictator will be rewarded more when choosing the fair allocation herself as compared to the case where the fair allocation is chosen by the delegate. The model also predicts that in the latter case the delegate is rewarded more than the dictator. One can say that the responsibility-shifting model predicts that the rewards in response to fair allocations are a mirror image of the punishments in response to unfair allocations.

Whether this symmetry (or mirroring) of rewards and punishments will hold empirically is less obvious than it may seem. There is evidence that punishment in response to “bad” behavior is stronger and more prevalent than rewards in response to “good” behavior (Offerman, 2002; Sefton et al., 2007; Croson and Konow, 2009; Kube et al., 2013). It cannot be ruled out that responsibility attribution also exhibits asymmetric patterns and causes the incentives to shift blame to be different from those to take credit. In fact, in political science, it is often argued that politicians and legislators have much stronger incentives to avoid blame than to claim credit. This is sometimes attributed to a “negativity bias” on the part of voters and public opinion which is rooted in the asymmetric treatment of gains and losses (Weaver, 1986; Pierson, 1994). Some scholars actually single out blame-shifting as a way to organize the entire field of public policy (Hood, 2010; Hinterleitner, 2018). Our experiment allows us to explore any asymmetric patterns of responsibility attribution at the individual level.

We find evidence in support of the responsibility attribution model in the presence of rewards. In our main treatment with rewards, the dictator receives a higher reward if the dictator directly chooses the fair allocation than in case the dictator delegates and the delegate chooses the fair allocation. Furthermore, the dictator receives a lower reward than the delegate in case the dictator delegates and the delegate chooses the fair allocation. Also on other accounts our responsibility model organizes the pattern of rewards well. For instance, the dictator receives the same reward when choosing the fair allocation directly as compared to delegating the decision to a random device which then happens to choose the fair allocation. This may seem remarkable, but it is in line with the assumption of Bartling and Fischbacher (2012) that only people and not nature can be held responsible for an outcome.

Several experimental studies show that individuals often choose to delegate decisions which may be considered unfair. This has been documented for allocation decisions in the dictator game (Hamman et al., 2010; Coffman, 2011; Bartling and Fischbacher, 2012; Choy et al., 2016; Gawn and Innes, 2019b) and the ultimatum game (Fershtman and Gneezy, 2001), but also for other types of unfavorable decisions such as communicating bad information (Garofalo and Rott, 2018), lying (Erat, 2013; Kandul and Kirchkamp, 2018; Gawn and Innes, 2019a), or bribing (Drugov et al., 2014). Gawn and Innes (2021) compare delegation decisions when delegating leads to a selfish allocation and when delegating is equivalent to a fair allocation in a dictator game. They find the delegation decision to be roughly 20 percentage points more frequent when it is payoff-equivalent to a selfish decision than in case it is payoff-equivalent to a more generous decision. This supports the “Machiavelli preference” to retain decision rights for generous decisions and delegate them for decisions that are selfish or harmful to others. In Gawn and Innes (2021) there is no role for blame or credit. In our paper, we use rewards by the recipients as an indicator of credit and responsibility attribution. We focus on recipients’ patterns of responses to dictators’ delegation and allocation decisions and examine whether these provide a foundation for Machiavelli’s advice.

A key finding in previous studies on delegation is that delegation is often effective in diverting punishment. For example, in dictator games with a punishment possibility, the principal receives a lower punishment in case an unfair allocation is chosen by a delegate rather than by the principal (Coffman, 2011; Bartling and Fischbacher, 2012; Oexl and Grossman, 2013). Similarly, in ultimatum games unfair offers are found to be accepted more frequently in case they are made by a delegate than in case they are directly made by the principal (Fershtman and Gneezy, 2001; Choy et al., 2016). In a collective decision game where three voters sequentially select between a fair and unfair allocation, Bartling et al. (2015) find that the voter whose decision is pivotal in choosing the unfair allocation receives the most punishment. Their finding suggests that subjects vote strategically to delegate the potentially unfavorable pivotal decision to subsequent voters.

Our paper contributes to the literature by examining reward behavior in response to the delegation of (fair) decisions. The literature has focused on punishments and there are few studies on rewards. One exception is Coffman (2011) who conducts an experiment in which participants can donate money to a charity to buy mosquito nets. Coffman (2011) finds that these donations are rewarded less by a third party when the intermediary role of the charity is made more salient. Although related, intermediation is different from delegation since a delegate is usually granted more decision autonomy. In Coffman (2011), the intermediary does not play an active role and cannot be rewarded, whereas we also study how delegation affects the rewards to the delegate.

Closer to our study, Eisenkopf and Fischbacher (2015) study a variant of the trust (investment) game in which the decision to trust can be delegated from one investor to another investor. In their design, delegation also generates an efficiency gain or loss which varies across treatments. Eisenkopf and Fischbacher (2015) hypothesize that the trustee will be more reciprocal toward delegation that increases efficiency than toward delegation that decreases efficiency. The results show that across treatments the investor who makes the actual investment decision is rewarded more by the trustee. Rewards, however, are unaffected by the (in)efficiency of the delegation decision indicating that trustees fail to display sophisticated behavior based on responsibility attribution as in Bartling and Fischbacher (2012). While Eisenkopf and Fischbacher (2015) study a trust game in which the investment decisions affect both efficiency and the payoff of the trustee, we follow Bartling and Fischbacher (2012) and focus on simple dictator games in which dictators’ decisions only have distributional consequences.

Our paper is also related to experimental papers studying responsibility attribution in gift-exchange games. These studies investigate the worker's effort provision in gift-exchange games when the wage-setting decision is made by the employer or delegated to either a neutral agent, a random device, or an agent whose payoff is related to the employer's. [Charness \(2000\)](#) finds that workers in a gift-exchange game respond to identical levels of wages with lower efforts when the wages are set by a neutral agent than when the wages are determined by a random device, suggesting a possible "responsibility-alleviation" effect. [Charness \(2004\)](#) and [Maximiano et al. \(2013\)](#) find that workers respond with lower efforts when a low wage is "intentionally" set by the employer than when it is delegated to either a random device or a neutral agent, which is similar to "punishing" unfavorable wages when they are directly made by the employer than when they are delegated. They also find that workers' effort provision in response to high levels of wages is higher when the wage is set by the employer than when it is delegated. Similar results are observed when the agent's payoff is proportional to the employer's payoff instead of being purely neutral ([Maximiano et al., 2013](#)). Unlike those articles, we explore the direct, monetary rewarding behavior of agents.

The rest of the paper is organized as follows. Section 2 explains the experimental design and procedural details. Section 3 predicts reward and punishment patterns based on our responsibility attribution model. Section 4 reports the behavioral patterns in reward and punishment as well as the delegation and allocation decisions. Section 5 contains a discussion of our findings. Section 6 provides concluding remarks.

2. Experimental design

In the experiment, we conduct a delegated dictator game with a similar design as in [Bartling and Fischbacher \(2012\)](#). There are three roles in the game: dictator (Player A), delegate (Player B), and recipient (Player C). Four players are matched together to form a group comprising one A, one B, and two Cs. A decides the allocation of 20 points among the four players by choosing between two different allocation options: (1) assigning 5 points to each player (the fair allocation) and (2) assigning 9 points each to A and B and 1 point each to both Cs (the unfair allocation). A has the option to delegate the allocation decision to B. If A chooses to delegate, B makes the allocation decision on behalf of A. The material interests of A and B are aligned and opposite to Cs'.

In the baseline treatment (*Baseline*), we implement the delegated dictator game without reward or punishment. Cs cannot make any decisions once A or B has chosen the allocation. In order to study responsibility attribution, we run a treatment with the option for Cs to reward A and/or B in the same delegated dictator game (*D&R*). The design of the *D&R* treatment is symmetric to the punishment set-up in [Bartling and Fischbacher \(2012\)](#): Cs are given the option to increase the payoffs of A and B by assigning extra points. They can choose to assign up to a total of seven points to either A, B, or both. One of the two reward plans is then randomly selected to be implemented.

There are two differences from the way punishment is implemented by [Bartling and Fischbacher \(2012\)](#), however. Both were aimed at encouraging the use of rewards directed at players A and B.¹ We wanted to prevent the rewarding rates from being too low to pick up any meaningful differences across conditions.² First, in [Bartling and Fischbacher \(2012\)](#) Cs pay a fixed cost of one point to buy seven punishment points. We decided to give Cs seven reward points for free. Unlike punishments in response to an unfair allocation, rewards in response to a fair allocation move the outcome away from an equal distribution, which can already be deemed costly even if not in monetary terms, in the presence of inequity aversion or some preferences for fairness. Second, to reduce possible demand effects ([Bartling and Fischbacher, 2012](#)) allow C players to punish not only A and B but also each other. In our experiment, however, we do not allow a C player to assign rewards to the other C player. Given that reward points come free, we wanted to rule out that Cs mostly reward each other at the cost of experimenters and would not have any points left to reward A and B.

In order to compare the reward of delegated decisions with that of pure allocation decisions, we include a treatment with the reward option but no delegation (*NoD&R*). In this treatment, A chooses between the two allocations, while B does not make any decision. After C players have decided whether and how to assign extra points to either A, B, or both, using the same rewarding mechanism as in the *D&R* treatment, one of them is randomly selected for implementation.

We conduct two additional variants of the *D&R* treatment: *Asymmetric* and *Random*, which mirror the ones in [Bartling and Fischbacher \(2012\)](#). In the *Asymmetric* treatment, if A chooses not to delegate, she can only choose the unfair allocation. If A delegates the decision, B can choose between the fair and the unfair allocations. In this treatment, delegation by A can unambiguously be interpreted as a "kind" action by A. In the *Random* treatment, instead of delegating to B, A can choose to delegate to a die which selects between the two allocations with the same probability as the Bs do in the *D&R* treatment. The goal of these two treatments is to (1) check whether other reward motives than responsibility attribution might be at play, and (2) offer a parallel comparison with the corresponding treatments with punishment in [Bartling and Fischbacher \(2012\)](#). The details are discussed in Section 3.

¹ [Bardsley \(2008\)](#) shows that in dictator games, the structure and size of participants' choice set affect their behavior. In the absence of costs, subjects also tend to have a preference for doing something, even if surplus-destroying, than doing nothing, see e.g. [Abbink and Sadrieh \(2009\)](#) or, for an extreme version, [Wilson et al. \(2012\)](#). Our design choices were thus likely to lead subjects to reward players rather than not. A design that would have included both punishments and rewards would have gone some way toward addressing that issue but would have allowed for less clear a comparison with [Bartling and Fischbacher \(2012\)](#).

² There are not many studies that compare the use of rewards and punishments. [Andreoni et al. \(2003\)](#) study versions of dictator games in which the recipient can punish, reward, or punish and reward. They find that recipients use both costly rewards and costly punishment, but that punishments are much more effective in reducing unfair offers. They also find that the possibility to punish reduces the use of rewards, while the possibility to reward does not reduce the use of punishments. [Sefton et al. \(2007\)](#) compare punishments and rewards in public goods games and find that deviations below the average of other group members were punished at a magnitude much higher than rewards for allocations above the average of other group members.

Table 1
Summary of treatments.

	Delegate option	C can reward	A can be fair
<i>Baseline</i>	To B	No	Yes
<i>D&R</i>	To B	Yes	Yes
<i>NoD&R</i>	No	Yes	Yes
<i>Asymmetric</i>	To B	Yes	No
<i>Random</i>	To die	Yes	Yes

Table 2
Number of subjects in each role.

	A	B	C	Total
<i>Baseline</i>	52	50	100	202
<i>D&R</i>	51	50	100	201
<i>NoD&R</i>	50	50	101	201
<i>Asymmetric</i>	50	50	100	200
<i>Random</i>	50	52	102	204

A summary of the different settings in all five treatments is shown in [Table 1](#).³

As in [Bartling and Fischbacher \(2012\)](#), the experiment was conducted using the strategy method. In each treatment, Bs have to decide which allocation to choose before they know whether A has chosen to delegate or not. Both Cs have to decide on how many reward points (if any) to assign for all possible scenarios before they know the decisions of A or B and before they know whether their decisions are selected to be implemented. In the *NoD&R* treatment, there are two possible scenarios: A choosing the fair allocation or A choosing the unfair allocation. In treatments with delegation (the *D&R*, *asymmetric*, and *random* treatments), there are up to four possible scenarios: A choosing the fair allocation and not delegating, A choosing the unfair allocation and not delegating, A choosing to delegate and B choosing the fair allocation, and A choosing to delegate and B choosing the unfair allocation. A summary of the number of subjects in each role in all five treatments is shown in [Table 2](#).⁴ At the end of the experiment, the decisions of one A, one B, and two Cs are randomly matched to form an A-B-2C group for payment calculations.⁵

The experiment was pre-registered at the AEA RCT Registry (AEARCTR-0006036) and approved by the Tilburg University School of Economics and Management Institution Review Board (IRB EXE 2020–011). The experiment was conducted as an online experiment on Prolific ([Palan and Schitter, 2018](#)) in June 2020. The *Baseline*, *D&R*, and *NoD&R* treatments were conducted first. The *asymmetric* and *random* treatments were conducted after we finished collecting data for the above treatments, so that the average probability of Bs choosing between the fair and the unfair allocation in the *D&R* treatment could be implemented for the die in the *Random* treatment. In order to keep the demographic features of subjects as similar as possible to those of [Bartling and Fischbacher \(2012\)](#) (who dealt with university students), we imposed the following restrictions on the subject pool: subjects' age should be between 18 and 30, subjects should be students, and the highest level of education completed should be at least high school. Moreover, we only recruited US residents as subjects. We took measures to minimize the gap in control between an online experiment and a lab experiment. To this effect, we implemented an additional recruitment criterion that the subject's past submissions on the Prolific platform should receive an approval rate of at least 90%.⁶ We kept our instructions as close as possible to those of [Bartling and Fischbacher \(2012\)](#), with modifications based on design differences and adjustments for the online implementation. Subjects were requested to complete the same set of practice questions as in [Bartling and Fischbacher \(2012\)](#) to improve their understanding of the instructions. The questions were changed into multiple-choice questions to adapt to online submissions. Subjects were only allowed to proceed if they chose the correct answer on the first attempt. The instructions for the three roles in the *D&R* treatment are shown in Section A.3 of the online Appendix. The experiment took around 15 min on average. The experimental points were converted to US dollars using the exchange rate of 5 points = \$1.⁷ In addition, each subject also received a fixed participation fee of \$1.1. For the *Baseline*, *D&R*, *D&P*, and *NoD&R* treatments, the average payment was around \$1.96. For the *asymmetric* and *random* treatments, the average payment was around \$2.03.

³ We also ran a treatment with punishment options identical to the one in [Bartling and Fischbacher \(2012\)](#). In this *D&P* treatment, C players can reduce the payoffs of the A player, the B player, or both. The punishment patterns we find replicate those in [Bartling and Fischbacher \(2012\)](#) and support the predictions of the responsibility attribution model. Details can be found in Section A.1 of the online Appendix.

⁴ Due to limitations in the randomization algorithm of the online questionnaire and the presence of control questions, the number of subjects in each role is not perfectly balanced. (That is, the total number of subjects in a given treatment need not be a multiple of 4.) This did not affect subjects' choices since they made decisions using the strategy method before any matching.

⁵ In each unbalanced treatment, there are one or two instances where a particular subject was matched twice to two different groups for payment purposes (not for decision-making purposes).

⁶ Submissions on the Prolific platform are rejected if the researcher has valid reasons indicating the subject was being negligent, e.g. the completion time was exceptionally short, crucial questions were skipped, or the subject failed attention checks.

⁷ The payment was transferred to the subjects via the Prolific system. The final payment was in British pounds using the exchange rate automatically adopted by the system.

Table 3
Theoretical responsibility of A and B in treatments with rewards.

	A fair	A delegates B/Die fair	
		$\alpha^+ + \delta\beta^+ < \beta^+$	$\alpha^+ + \delta\beta^+ > \beta^+$
<i>D&R</i>			
Responsibility of A	1	R_{A1}^+	0
Responsibility of B	0	R_{B1}^+	1
<i>Asymmetric</i>			
Responsibility of A	–	R_{A2}^+	0
Responsibility of B	–	R_{B2}^+	1
<i>Random</i>			
Responsibility of A	1		1
Responsibility of B	0		0

Note: $R_{B1}^+ > R_{A1}^+$ if $\beta^+ < (1 + \alpha^+)/(2 - \delta)$; $R_{B2}^+ > R_{A2}^+$ if $\beta^+ < 1/(2 - \delta)$; $R_{A2}^+ > R_{A1}^+$; $R_{B2}^+ < R_{B1}^+$.

3. Hypotheses

Our primary interest lies in the reward behavior of the C players. Although there are several ways to interpret and model responsibility attribution, we base our hypotheses on the measure of responsibility proposed by [Bartling and Fischbacher \(2012\)](#). The measure assigns most responsibility for an outcome to a player whose actions had the largest impact on the probability that the outcome occurred. [Bartling and Fischbacher \(2012\)](#) find that this measure is a good predictor for the punishments that C players administer to A and B players in case of an unfair outcome. We hypothesize that it also predicts the rewards that C players assign to A and B players in case of a fair outcome. Moreover, we hypothesize that C players will not reward A or B in case of an unfair outcome.

In the *NoD&R* treatment, A is the only player who affects the allocation, and A is assigned full responsibility for a fair outcome. In the other three treatments with delegation and reward (*D&R*, *Asymmetric*, and *Random*), a player (A and B) who is perceived to have increased the probability that the fair allocation is chosen will be held responsible and will receive a corresponding reward. The stronger the responsibility, the higher the reward.

More precisely, let C's belief about the probability that A and B choose the fair allocation be denoted by α^+ and β^+ , respectively, and let the belief about the probability that A delegates be denoted by δ . In the *D&R* treatment, the *ex ante* probability of the fair allocation is $\alpha^+ + \delta\beta^+$. If A delegates, the *ex post* probability of the fair allocation is β^+ . If $\alpha^+ + \delta\beta^+ < \beta^+$, both A and B are responsible for increasing the probability. A's share of contribution to the probability increase is $R_{A1}^+ = (\beta^+ - \alpha^+ - \delta\beta^+)/(\beta^+ - \alpha^+ - \delta\beta^+)$ and B's share $R_{B1}^+ = (1 - \beta^+)/(\beta^+ - \alpha^+ - \delta\beta^+)$. If $\beta^+ < (1 + \alpha^+)/(2 - \delta)$, B has a larger impact on the probability increase than A, and *vice versa*. If $\alpha^+ + \delta\beta^+ > \beta^+$, A reduces the probability of a fair outcome by delegating and should not be held responsible, while B is fully responsible since B becomes the only one increasing the probability of the fair outcome.

In the *Asymmetric* treatment, since A can only choose the unfair allocation if she does not delegate, the *ex ante* probability of the fair outcome is $\delta\beta^+$. If A delegates and B chooses the fair allocation, A's share in the probability increase is $R_{A2}^+ = (\beta^+ - \delta\beta^+)/(\beta^+ - \delta\beta^+)$ and B's responsibility is $R_{B2}^+ = (1 - \beta^+)/(\beta^+ - \delta\beta^+)$. If $\beta^+ < 1/(2 - \delta)$, B is more responsible for the probability increase than A, and *vice versa*. A's responsibility is always no less than the corresponding case in the *D&R* treatment, while B's responsibility is always no more than the corresponding case in the *D&R* treatment. In the *Random* treatment, following [Bartling and Fischbacher \(2012\)](#)'s assumption that an individual's responsibility is not affected by moves of nature, if A delegates to the die which results in the fair allocation, only A takes full responsibility. The predicted responsibility of A and B in the three treatments with delegation and reward as represented by their influence on the probability of the fair outcome is shown in [Table 3](#).

Let $r_i(j, x)$ denote the reward for i in the *D&R* treatment when the direct allocator who chooses the allocation is j and the chosen allocation is x , with $i, j \in \{A, B\}$, $x \in \{f, u\}$. Based on how Cs attribute responsibility for the fair outcome, we have the following predictions about the reward pattern.

Hypothesis 1. $r_A(\cdot, u) = r_B(\cdot, u) = 0$. The reward for both players is zero when the outcome is unfair, independently of who made the decision.

Hypothesis 2. $r_A(A, f) > r_A(B, f)$, $r_B(B, f) > r_B(A, f)$. When the chosen allocation is fair, player i receives a higher reward in case she is the direct allocator than in case she is not.

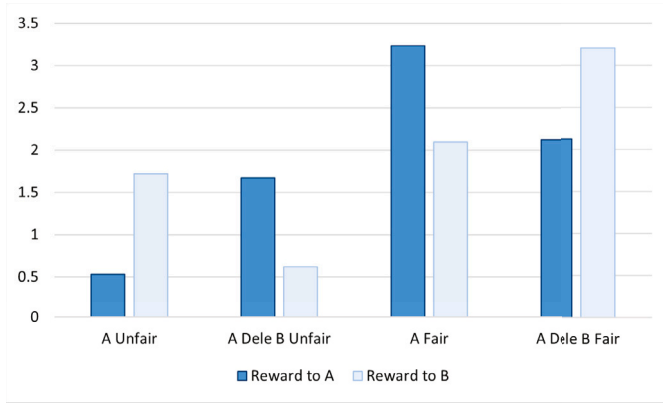
Hypothesis 3.

1. $r_A(A, f) > r_B(A, f) = 0$. If A does not delegate and directly selects the fair allocation, only A is rewarded.
2. $r_A(B, f) \geq 0$, $r_B(B, f) > 0$; $r_A(B, f) > r_B(B, f)$ if $\alpha^+ + \delta\beta^+ < \beta^+$ and $\beta^+ > (1 + \alpha^+)/(2 - \delta)$, $r_B(B, f) > r_A(B, f)$ otherwise. If A delegates and B selects the fair allocation, B receives a higher reward than A for some combinations of probabilities.
3. $r_A(B, f) \geq r_B(A, f)$, equality holds if $\alpha^+ + \delta\beta^+ \geq \beta^+$. Whether delegation completely reduces the reward for A depends on the specific combinations of probabilities.

Table 4
Average reward for A and B in treatments with reward.

	A unfair		B/die unfair		A fair		B/die fair	
	To A	To B	To A	To B	To A	To B	To A	To B
<i>D&R</i>	0.53 (1.19)	1.67 (2.33)	1.67 (2.21)	0.57 (1.17)	3.24 (1.90)	2.05 (1.43)	2.13 (1.36)	3.16 (1.86)
<i>NoD&R</i>	0.63 (1.23)	2.45 (2.78)	–	–	3.36 (1.52)	2.28 (1.24)	–	–
<i>Asymmetric</i>	0.59 (1.22)	2.01 (2.51)	2.36 (2.65)	0.81 (1.45)	–	–	2.78 (1.26)	3.1 (1.36)
<i>Random</i>	1.01 (1.71)	1.87 (2.42)	1.24 (1.71)	1.79 (2.18)	3.01 (1.95)	1.89 (1.39)	2.76 (1.66)	2.56 (1.56)

Note: Column 2 (“B/die unfair”) and Column 4 (“B/die fair”) refers to scenarios in which A delegates and B makes the allocation decision. In Column 2 and Column 4 of the last two rows, average reward when A delegates and the die gives the unfair/fair outcome is shown respectively. Standard errors are shown in parenthesis.



(a) *D&R* treatment



(b) *NoD&R* treatment

Fig. 1. Average reward in *NoD&R* and *D&R* treatment.

Let $r_i^{AS}(\cdot, \cdot)$ denote the reward for player i in the *Asymmetric* treatment, and $r_i^{RD}(\cdot, \cdot)$ denote the reward for player i in the *Random* treatment. We have the following additional predictions comparing the three treatments.

Hypothesis 4. $r_A^{RD}(A, f) = r_A^{RD}(\text{die}, f) > 0$, $r_B^{RD}(A, f) = r_B^{RD}(\text{die}, f) = 0$. In the random treatment, delegating to a die does not affect the reward for any of the players.

Hypothesis 5. $r_A^{RD}(A, f) = r_A^{RD}(\text{die}, f) > r_A^{AS}(B, f) \geq r_A(B, f) \geq 0$, the last two equality hold if $\alpha^+ + \delta\beta^+ \geq \beta^+$; $r_B(B, f) \geq r_B^{AS}(B, f) > r_B^{RD}(B, f) = 0$, the first equality holds if $\alpha^+ + \delta\beta^+ \geq \beta^+$. If A delegates and the outcome is fair, A's reward in the *Random* treatment is the highest among all three treatments. A's reward in the *Asymmetric* treatment is no less than in the *D&R* treatment. B's reward in the *D&R* treatment is no less than in the *Asymmetric* treatment.

Hypotheses 3 and 5 depend on C-subjects' beliefs regarding the actions taken by other players. In our analysis, we will use the aggregate empirical frequencies of decisions, at the treatment level, to stand for those beliefs.

We note that the attribution of responsibility for an outcome to players is not the only possible motive for rewards and punishments. Other motives include a preference for an equal payoff distribution (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000), or a preference to reciprocate behavior which is perceived to be (un)kind (Rabin, 1993; Charness and Rabin, 2002; Dufwenberg and Kirchsteiger, 2004; Falk and Fischbacher, 2006). In some cases the predictions of these alternative models overlap with those of our responsibility attribution model; in other cases, they conflict. We will point out specific instances when we go over the results.

4. Results

4.1. Reward behavior

We summarize the average reward for each player under all four possible scenarios in each treatment with reward in Table 4. Fig. 1 shows the average reward for A and B in all possible scenarios in the *D&R* treatment (Fig. 1(a)) and in the no-delegation

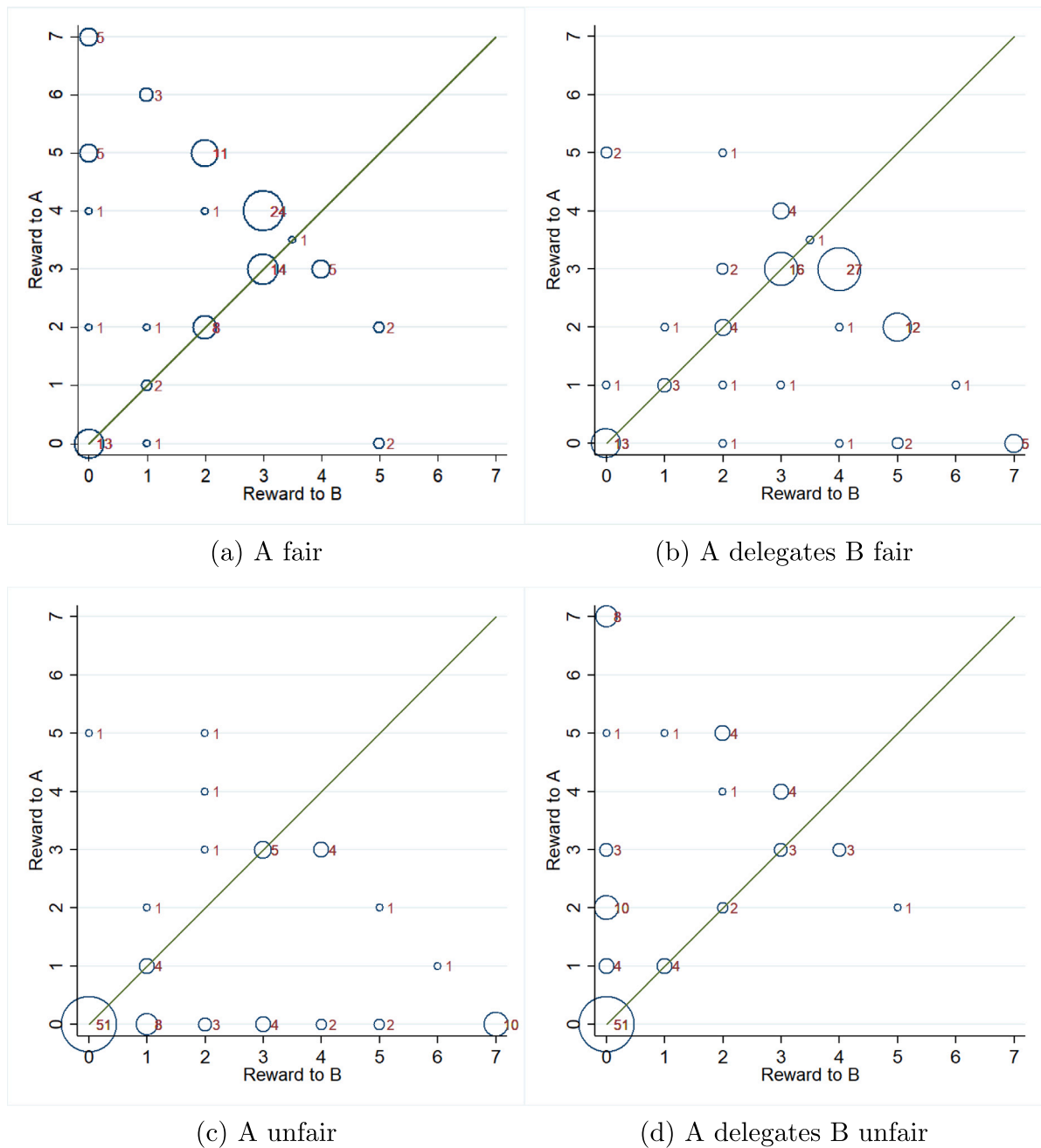


Fig. 2. Distribution of individual rewards in D&R treatment.

benchmark *NoD&R* treatment (Fig. 1(b)). Fig. 2 provides more detailed information about individual reward decisions in the four scenarios in the *D&R* treatment. Fig. 3 shows the average reward for A and B in all possible scenarios in the *Asymmetric* treatment (Fig. 3(a)) and the *Random* treatment (Fig. 3(b)).

From Table 4, one sees that the reward for both A and B is typically higher after a fair allocation than after an unfair allocation. The only exceptions are the rewards for B in the *NoD&R* treatment and the *Random* treatment. At the same time, in all four treatments, and contrary to Hypothesis 1, both A and B receive a positive reward in each of the four possible scenarios, that is, even if the allocation is unfair and irrespective of whether A or B is the direct allocator. This is not in line with our responsibility attribution model, which predicts a reward to occur only when the outcome is fair and when a player has affected the likelihood of the fair outcome. We will come back to this in Section 5.

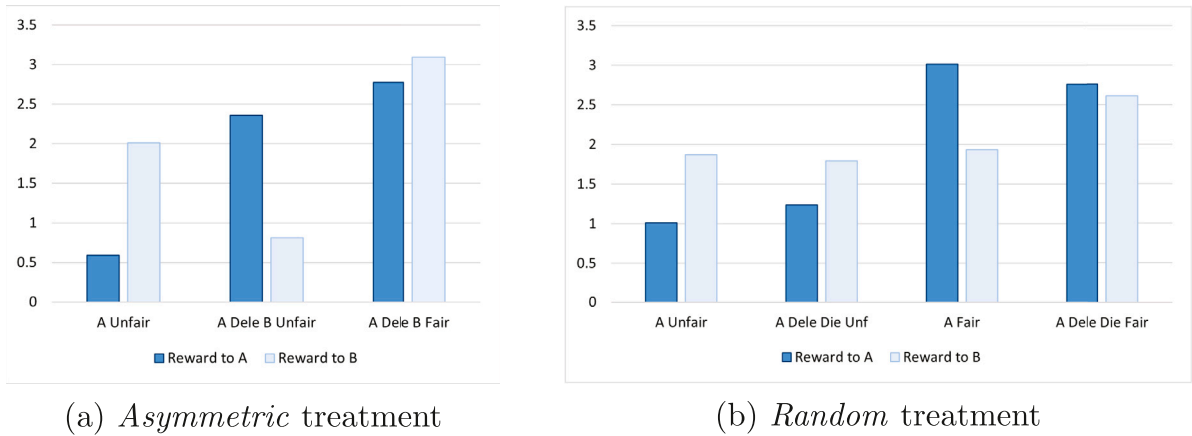
Fig. 3. Average reward in *Asymmetric* and *Random* treatment.

Table 5

Delegation and allocation decisions of A and B.

	Unfair (%)	Fair (%)	Delegate (%)	Observations
<i>Baseline</i>				
A	31	60	10	52
B	34	66	–	50
<i>D&R</i>				
A	31	53	16	51
B	34	66	–	50
<i>NoD&R</i>				
A	28	72	–	50
<i>Asymmetric</i>				
A	28	–	72	50
B	24	76	–	50
<i>Random</i>				
A	28	56	16	50

Result 1. $r_A(\cdot, u) > 0$, $r_B(\cdot, u) > 0$. Both players receive a positive reward if the chosen allocation is unfair, regardless of who is the direct allocator.

This being said, our *D&R* treatment shows a reward pattern across outcomes which is in line with our responsibility attribution model. As predicted by [Hypothesis 2](#), when the outcome is fair, a player receives a significantly higher reward when she is the direct allocator than when she is not. As shown in the last two columns of the first panel in [Table 4](#), A receives a reward of 3.24 when A directly selects the fair allocation, while she only receives a reward of 2.13 when she delegates and the fair allocation is chosen by B. Similarly, B's reward is 2.05 when A directly selects the fair allocation, while it is 3.16 when A delegates and B chooses the fair allocation. Both pairwise comparisons are statistically significant, supported by a two-sided Wilcoxon signed-rank tests with $p < 0.01$.

Result 2. $r_A(A, f) > r_A(B, f)$, $r_B(B, f) > r_B(A, f)$. When the outcome is fair, player i receives higher rewards when she is the direct allocator than when she is not.

The comparisons of rewards between A and B in the *D&R* treatment also exhibit a pattern consistent with [Hypothesis 3](#). If A directly selects the fair allocation, A's reward (3.24) is significantly higher than B's reward (2.05); if A delegates and B selects the fair allocation, B's reward (3.16) is also significantly higher than A's reward (2.13) ($p < 0.01$ in a two-sided Wilcoxon signed-rank test for both comparisons). With our observed delegation and allocation decisions of A and B as shown in [Table 5](#) in Section 4.2, we compute the frequency of A being fair to be $\hat{\alpha}^+ = 0.63$, the frequency of B being fair to be $\hat{\beta}^+ = 0.66$, and the frequency of delegation to be $\hat{\delta} = 0.16$. Since $\hat{\alpha}^+ + \hat{\delta}\hat{\beta}^+ = 0.73 > \hat{\beta}^+$, if we assume that C-subjects' beliefs are consistent with empirical frequencies, then the comparisons of the corresponding rewards for A and B are consistent with the second point of [Hypothesis 3](#).

We find that A's average reward (2.13) when A delegates and B selects the fair allocation is not significantly different from B's average reward (2.05) when A directly selects the fair allocation ($p = 0.94$ in a two-sided Wilcoxon signed-rank test). Similarly, A's average reward (3.24) when A directly selects the fair allocation is also similar with the reward for B (3.16) when B is the direct allocator for the fair allocation. This pattern can be seen more clearly in [Fig. 1\(a\)](#). The reward patterns for A and B when A direct selects the fair allocation mirror the reward patterns for B and A when A delegates and B selects the fair allocation. With our estimated $\hat{\alpha}^+$, $\hat{\beta}^+$, and $\hat{\delta}$, since $\hat{\alpha}^+ + \hat{\delta}\hat{\beta}^+ = 0.73 > \hat{\beta}^+$, these comparisons are consistent with the third point of [Hypothesis 3](#).

Result 3.

1. $r_A(A, f) > r_B(A, f)$. If A does not delegate and directly selects the fair allocation, A receives more reward than B.
2. $r_B(B, f) > r_A(B, f)$. If A delegates and B selects the fair allocation, B receives more reward than A.
3. $r_A(B, f) = r_B(A, f)$. Given the allocation result, the indirect allocator in the two different scenarios receives the same level of reward.

To summarize our findings in the D&R treatment, when looking at pairwise comparisons, we find confirming evidence for the predictions based on our responsibility attribution model. However, in that treatment, A and B subjects receive a positive reward even when the outcome is unfair, and B receives a positive reward when A does not delegate and B does not make any (consequential) decision.

That said, in the D&R treatment, these rewards when the outcome is unfair also exhibit a clear and intuitive pattern: (1) $r_A(A, u) < r_A(B, u)$, $r_B(B, u) < r_B(A, u)$. When the outcome is unfair, a player receives a significantly lower reward when she is the direct allocator than when she is not. (2) $r_A(A, u) < r_B(A, u)$. If A does not delegate and directly selects the unfair allocation, A receives less reward than B. (3) $r_B(B, u) < r_A(B, u)$. If A delegates and B selects the unfair allocation, B receives less reward than A. (4) $r_A(B, u) = r_B(A, u)$. When the outcome is unfair, we also find that the indirect allocator receives similar levels of reward. The rewards when the outcome is unfair exhibit a mirror image to the reward pattern when the outcome is fair, as summarized in Results 2 and 3.

To rule out that these average patterns result from some coincidental aggregation of heterogeneous decisions, we also have a brief look at the distribution of the individual reward decisions. Fig. 2 shows the distribution of each individual C's reward assignment decisions in the D&R treatment. The upper two panels show the two scenarios when the fair allocation is chosen. The horizontal axis denotes the reward points assigned to B and the vertical axis denotes the reward points to A. In the upper two panels, a number of reward assignments fall on the 45 degree line, indicating that a fraction of Cs assign the same level of reward to both A and B as long as the outcome is fair, regardless of who directly makes the final allocation decision. Other than that, most Cs assign more rewards to A than to B when A directly selects the fair allocation (Panel 2(a)) and more rewards to B than to A when A delegates and B selects the unfair allocation (Panel 2(b)). This confirms that the majority of Cs' decisions when the chosen allocation is fair are in line with the predictions of our responsibility attribution model: the direct decision-maker receives a higher reward.

The lower two panels show the two scenarios when the chosen allocation is unfair. Either when A directly selects the unfair allocation (Panel 2(c)) or when A delegates and B selects the unfair outcome (Panel 2(d)), around half of the Cs assign zero reward to both A and B. However, another half of Cs still assign a positive reward even when the unfair allocation is chosen. In Panel 2(c), when A does not delegate and directly selects the unfair allocation, a considerable fraction of Cs only assigns a positive reward to B, as can be seen on the horizontal axis. This can be regarded as a "bystander" reward to B for not contributing to the unfair outcome. In Panel 2(d), when A delegates and B chooses the unfair allocation, a large fraction of Cs only rewards A with a positive amount, as shown on the vertical axis. Apart from the "bystander" reward motive, a C player might also reward A if C believes that A could reasonably expect B to choose the fair allocation upon delegation.

Given the high number of choices to be made by subjects, it is hard to classify them by individual types. Among the 100 Cs in the D&R treatment, we nonetheless identify a few prominent types, based on their reward patterns across the four scenarios. 8 always assign 0 reward to both A and B in all four scenarios ($r_A(A, f) = r_B(A, f) = r_A(B, f) = r_B(B, f) = r_A(A, u) = r_B(A, u) = r_A(B, u) = r_B(B, u) = 0$). Among the other 92 Cs who assign a positive reward to at least one player in at least one scenario, 45 assign reward with some responsibility-shifting concerns when the outcome is fair, assigning a higher reward to the direct allocator who directly selects the fair allocation ($r_A(A, f) > r_B(A, f)$ and $r_A(B, f) < r_B(B, f)$). Among these 45 Cs, 13 assign 0 reward when the outcome is unfair ($r_A(A, u) = r_B(A, u) = r_A(B, u) = r_B(B, u) = 0$), while 22 exhibit a reward pattern in the two unfair scenarios that mirrors that in the two fair scenarios, assigning a lower reward to the direct allocator who directly selects the unfair allocation ($r_A(A, u) < r_B(A, u)$ and $r_A(B, u) > r_B(B, u)$). In addition, 22 Cs do not distinguish the reward to A and to B in each scenario ($r_A(A, f) = r_B(A, f)$, $r_A(B, f) = r_B(B, f)$, $r_A(A, u) = r_B(A, u)$, and $r_A(B, u) = r_B(B, u)$), among which 15 assign 0 reward in the two unfair scenarios ($r_A(\cdot, u) = 0$) and a positive reward ($r_A(\cdot, f) > 0$) in at least one of the two fair scenarios. Thus, it is not the case that every subject follows the predictions of the responsibility attribution model. A significant minority of subjects uniformly reward As and Bs (if at all). Still, sufficiently many subjects follow some (if not all) the predictions of the model and generate the aggregate pattern we have reported about.

In order to further explore the predictions of the responsibility attribution model, we now turn to the additional treatments we implemented. In the *Asymmetric* treatment (Fig. 3(a)) and the *Random* treatment (Fig. 3(b)), we observe a pattern reminiscent of the one in D&R treatment, as well as some new findings. In the *Asymmetric* treatment, each player receives a significantly lower reward when she directly selects the unfair allocation than when she is not the direct allocator ($p < 0.01$ in one-sided Wilcoxon signed-rank tests). The reward for A and B when A directly selects the unfair allocation mirrors the reward for B and A when A delegates and B selects the unfair allocation. A's reward when A delegates and B selects the unfair allocation (2.36) is slightly higher than the reward for B when A directly selects the unfair allocation (2.01), but the difference is not statistically significant ($p = 0.18$ in a Wilcoxon signed-rank test). Comparing the *Asymmetric* treatment with the D&R treatment, A receives 2.78 points of reward when A delegates and B selects the fair allocation in the *Asymmetric* treatment, which is significantly higher than in the D&R treatment (2.13) ($p < 0.01$ in a one-sided Mann-Whitney u test). A's reward (2.36) when A delegates and B selects the unfair allocation in the *Asymmetric* treatment is also mildly significantly higher ($p < 0.1$ in a one-sided Mann-Whitney u test) than in the D&R treatment (1.67). This points to the fact that A's action space tends to affect the reward she receives. Cs take into account the fact that delegating is the only possibility to reach the fair outcome and raise the reward for A if A delegates. This pattern is predicted by our responsibility attribution model. It is also compatible with intention-based reciprocity. However, it clearly violates any outcome-based theory of retribution.

Result 4. $r_A^{AS}(B, \cdot) > r_A^{AS}(A, u)$, $r_A^{AS}(B, \cdot) > r_A(B, \cdot)$. In the Asymmetric treatment, delegation increases A's reward, compared with the case when A does not delegate.

In the *Random* treatment, A receives a significantly higher reward when the outcome is fair than when the outcome is unfair, both when A directly makes the allocation decision and when A delegates ($p < 0.01$ in one-sided Wilcoxon signed-rank tests). Being the direct allocator does not have a significant impact on the level of rewards in the *Random* treatment. When the final allocation is unfair, the reward for A does not differ significantly if A directly makes the decision (1.01) or if A delegates the decision (1.24) ($p = 0.16$ in a two-sided Wilcoxon signed-rank test). When the final allocation is fair, the reward for A if A delegates and B selects the fair option is as high as 2.76, also not significantly different from A's reward (3.01) if A directly selects the fair option ($p = 0.22$ in a two-sided Wilcoxon signed-rank test). In addition, when A delegates and the fair allocation is subsequently selected, A receives a significantly higher reward in the *Random* treatment (2.76) than in the *D&R* treatment (2.13) ($p < 0.01$ in a one-sided Mann-Whitney u test). This provides confirming evidence for our prediction that delegating to a random device does not dilute A's responsibility.

Result 5. $r_A^{RD}(A, x) = r_A^{RD}(\text{die}, x)$, $x = u, f$. Given the allocation result, delegating to a random device does not have a significant impact on the level of reward for A.

4.2. Delegation and allocation decisions

Table 5 summarizes the delegation and allocation decisions of A and B in all treatments. A large proportion of both As and Bs choose the fair allocation, which is consistent with the common finding from previous dictator games (Engel, 2011; Grech et al., 2022). Compared with previous binary choice dictator games, our subjects exhibit a higher degree of generosity, though. In the treatments where B is asked to choose between the two allocations, the fraction of Bs who choose the fair allocation ranges between 66% and 76%. In the treatments where As are asked to choose among the two allocations and delegation, the fraction of As who select the fair allocation ranges between 44% and 60%.

We do not find a significant effect of adding a reward option on the delegation and allocation decisions. Around 10% of As choose to delegate in the *Baseline* treatment, and this fraction increases to 16% in the *D&R* treatment. However, this difference is not significant ($p = 0.62$ with a Fisher's exact test). Compared with the delegation rate of 17% in the no punishment treatment and 55% in the *D&P* treatment of Bartling and Fischbacher (2012), the rate of delegation is relatively low in our experiment. It is also low compared to other studies on delegated dictator games (38% in Oexl and Grossman (2013), 40% in Hamman et al. (2010), and 22.2% in Gawn and Innes (2019b)) and delegated ultimatum games (73% in Fershtman and Gneezy (2001) and 38% in Choy et al. (2016)).

In the absence of sanction possibilities (punishments or rewards), the delegation option is mainly used by A players who hope that the B will choose the unfair option but who are unwilling to do so themselves.⁸ Moreover, these A players often get their way as a large majority of Bs does indeed choose unfair in Bartling and Fischbacher (2012). In our experiments, however, the rate at which players choose fair is substantially higher than in Bartling and Fischbacher (2012). This holds for both A and B players. For instance, the rate at which B players choose fair in our *Baseline* treatment (66%) is much higher than in the comparable *D&noP* treatment in Bartling and Fischbacher (2012) (18%). In view of this, it is not surprising that delegation is less prevalent in our experiment than in those of Bartling and Fischbacher (2012) and others. It looks like we have a larger fraction of A players with a preference for a fair outcome. And even those that do not, are less likely to attain an unfair outcome by delegating the decision to the B player. So, they might just as well choose fair themselves and have a shot at a higher reward.

In our treatments with rewards, the low level of delegation is consistent with our finding that delegation decreases A's reward when the outcome is fair and that the majority of A and B choose the fair allocation. Still, on the basis of treatment averages, in *D&R*, A pockets 8.24 if she directly chooses the fair allocation, while she can expect about 8.31 if she delegates. The same applies to the *Random* treatment, where A's expected payoff is 8.01 if she chooses the fair allocation and 8.58 if she delegates. However, in all treatments, directly selecting the unfair allocation gives the highest expected payoff for A (9.53 in *D&R*, 9.59 in *Asymmetric*, 10.01 in *Random*). Thus, it does not look as if the majority of subjects in our experiment were maximizing their expected payoff (as, of course, is often found in dictator games). We stress, however, that, in reality, subjects did not know the various empirical frequencies with which A, B and C subjects make their decisions.

In the *NoD&R* treatment where there is no delegation option and A can only choose between the two allocations, around 72% of A chooses the fair allocation. In the *Asymmetric* treatment where A can only choose between directly selecting the unfair allocation and delegating, around 72% of A choose to delegate. This amount coincides with the sum of the fractions of As who directly choose the fair allocation and who choose to delegate in the three treatments where the delegation option is available (i.e. *Baseline*, *D&R*, and *Random*). This is therefore consistent with the possibility that dictators who delegate are those who would choose the fair allocation if that were an option. A similar pattern is observed when comparing a direct dictator game and a delegated dictator game in Gawn and Innes (2019b).

⁸ This follows from the fact that the introduction of the delegation possibility in Bartling and Fischbacher (2012) does not reduce the percentage of As who choose unfair, and only reduces the percentage of As who choose fair. So As who choose to delegate would have chosen fair in the absence of delegation.

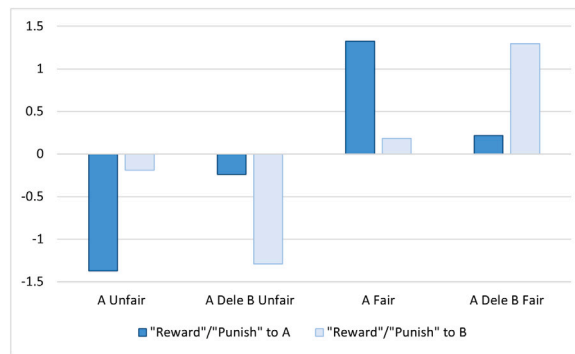


Fig. 4. Adjusted “reward” and “punishment” in the D&R treatment.

5. Discussion

Our findings offer considerable support for the predictions of our responsibility attribution model. In the *D&R* treatment, the direct allocator receives more reward for a fair outcome. Delegation followed by a fair choice by B leads to a higher reward for B, but A is still held partially responsible for the outcome. The fact that the reward pattern is affected by the change in A’s strategy set in our *Asymmetric* treatment and, in particular, the fact that A gets a higher reward when delegating to an unfair B in that treatment than in the *D&R* treatment, is consistent with both responsibility attribution and intention-based reciprocity. However, our observation that the reward for A is similar whether A makes the decision herself or delegates to a random die in the *Random* treatment provides supporting evidence for the responsibility-attribution argument.

One notable deviation from the predictions of the responsibility model is that, on average, both player A and player B receive positive rewards even in case the outcome is unfair (Result 1). Inequality aversion or fairness preferences do not offer an explanation for this as rewards only distort the outcome further away from equality. Intention-based reciprocity also cannot account for positive rewards in response to an unfair allocation chosen by A or B. A preference for efficiency may explain the use of rewards since rewards do indeed increase total payoffs. Surely the fact that rewards are costless in our experiment may have contributed to their use in this respect.⁹ Still, the level of rewards in response to unfair allocations is not uniform across conditions. There are some noteworthy patterns which we will now explore in more detail.¹⁰

One interesting observation is that B receives a significantly higher reward than A when A directly selects the unfair allocation. When A does not delegate, B is a pure bystander who does not make any (consequential) decision. This seems to indicate that B is rewarded as a bystander for doing nothing when A selects the unfair allocation. This cannot be explained by intention-based reciprocity (B does not do anything) or outcome-based social preferences (B already enjoys a high payoff when the unfair allocation is chosen and rewarding him makes the allocation even more unequal). It seems as if B is rewarded for *not* being (held) responsible for an unfavorable outcome.

A possible interpretation is that Cs use the different levels of reward as means to implement *both* rewards and punishments. By taking a baseline reward level as a reference point, a reward below this level can be regarded as a punishment while a reward above this level can be regarded as a genuine reward.¹¹ A possible candidate for such a baseline level is the average reward for B when A does not delegate. Averaged over the cases that A implements a fair or unfair allocation this baseline reward level is 1.91 in our *D&R* treatment.¹² Fig. 4 shows the adjusted “punishment” with the unfair outcome and the adjusted “reward” with the fair outcome for each player, subtracting the reference value 1.91 from the average reward as shown in Fig. 1(a). One can see that the direct allocator receives a higher “reward” than the other player when the outcome is fair and is more severely “punished” than the other player when the outcome is unfair. The “reward” patterns with the fair outcome are a mirror image of the “punishment” patterns with the unfair outcome. These results indicate that when using the different levels of reward for both punishing and rewarding, delegation reduces both punishment and reward in line with our responsibility attribution model.

To further explore this possibility, we can again use the responsibility attribution model to generate predictions for such rewards and “punishments”. Responsibility for the fair allocation (ρ^+) depends on each player’s contribution to the increase in the probability of the fair allocation being selected, while responsibility for the unfair allocation (ρ^-) depends on each player’s contribution to the increase in the probability of the unfair allocation being selected. We calculate the values for ρ^+ and ρ^- according to the observed

⁹ It is notable though that Andreoni et al. (2003) also find positive costly rewarding in response to unfair offers.

¹⁰ It should be noted that the analysis in this section is exploratory and was not pre-registered.

¹¹ This is very similar to Nikiforakis and Mitchell (2014) who find that unaffected third-parties withhold rewards to signal their disapproval with unfair proposals by a dictator.

¹² Admittedly, the selection of this baseline level is quite arbitrary. Another possible candidate would be the average reward for B in the *NoD&R* treatment. Choosing alternative baseline reward levels does not alter the main gist of our exercise, however.

Table 6
Regression of reward on responsibility.

	(1)	(2)
ρ^+	1.109*** (0.103)	0.947*** (0.112)
ρ^-	-1.314*** (0.0894)	-1.204*** (0.118)
Fair		0.256** (0.117)
Constant	2.095*** (0.0792)	1.985*** (0.115)
Observations	2620	2620
Number of Subjects	403	403

Note: Dependent variable is C's each reward assignment decision. This table presents regression results using a random-effect model. Random effects are clustered by each individual. Standard errors are shown in parenthesis. ***p < 0.01, **p < 0.05, *p < 0.1.

share of A and B's choices of fair, unfair, and delegation decisions in each treatment. The calculations can be found in Section A.2 of the online Appendix.

We run regressions of C's individual rewards for a player (A or B) on the responsibility measures, pooled across fair and unfair outcomes and pooled across treatments. The results are shown in Table 6. Rewards correlate positively with the responsibility of a player (A or B) for a fair outcome (ρ^+), and negatively with the responsibility for an unfair outcome (ρ^-). The constant term can be interpreted as the baseline level of reward we discussed above. By construction, the responsibility variables ρ^+ and ρ^- are correlated with whether the outcome is fair or unfair. Controlling for the outcome, the results in column (2) show that this reward pattern is robust.¹³

It is intuitive and appealing to think of C-players as using a baseline level of reward to be able to administer both (extra) rewards and punishments. However, this interpretation is speculative and we stress that our experiment was not designed specifically to test that hypothesis.

6. Conclusion

We conduct online experiments implementing a delegated dictator game where recipients can reward the principal or the delegate (or both). We find that the dictator receives a higher reward when directly choosing the fair allocation than when the delegate does it on her behalf (and conversely). In that sense, delegation in the presence of rewards dilutes credit-taking in the same way as delegation in the presence of punishments dilutes blame-taking. The predictions of the extension of a simple responsibility attribution model put forward by Bartling and Fischbacher (2012) are confirmed, with one exception: rewards are systematically given, even when the outcome is unfair and even to players who do not make any decision and thereby do not contribute to the realization of the outcome.

We find, however, that, once low and high rewards are seen as deviations from a baseline level and re-interpreted as actual punishments and actual rewards, respectively, delegation reduces both rewards and punishments for the dictator in the case of a fair or unfair outcome, respectively.

In line with our responsibility attribution model, recipients also take the action space of the principal into account when assigning rewards and delegating to a random device does not produce a distancing effect. Our results on delegation decisions are consistent with the possibility that subjects who delegate are those who would be fair in the absence of the delegation option.

Our results indicate that delegation reduces the credit associated with positive outcomes and the blame associated with negative outcomes by shifting the responsibility from the indirect allocator to the direct allocator. However, the credit-alleviation and the blame-shifting effect of delegation do not lead dictators strategically to use delegation in our experiment.

We observe a lower proportion of delegation decisions compared with previous studies with delegated allocation games and essentially no change in the delegation rate in the presence of rewards or punishments. The high proportion of subjects directly choosing the fair allocation in our environment may play a role. However, we think this calls for further investigation of the robustness of the strategic blame-shifting delegation phenomenon. In our view, the fact that subjects may use a reward system as a way to avail themselves the possibility of (indirectly) punishing other players should also be investigated as a matter of priority.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

¹³ A regression including additional controls (demographic variables such as age, gender, and education) is shown in Table A6 in Section A.2 of the online Appendix. It does not affect the main results.

Data availability

Data will be made available on request

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.euroecorev.2023.104522>.

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